

Cross-Market Intraday Time-Series Momentum

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1. Introduction

The momentum effect, a key market anomaly which has so far no consensus theoretical explanations, is found to exist throughout global financial markets, and the momentum factor is often considered as a predictor of future returns (Jegadeesh and Titman, 1993; Fama and French, 1996, 2017; Carhart, 1997; Hong and Stein, 1999; Moskowitz and Grinblatt, 1999; Jegadeesh and Titman, 2001; Moskowitz et al., 2012). Though the traditional finance school and the behavioural finance school differ in their approach to explain the momentum effect, they concur on the profitability of financial market strategies developed based on the momentum effect (Barberis et al., 1998; Conrad and Kaul, 1998; Daniel et al., 1998; Hong and Stein, 1999; Hong et al., 2000; Okunev and White, 2003; Miffre and Rallis, 2007; Vanstone et al., 2017; Ashour et al., 2022).

The previous momentum research is often limited to monthly or weekly return patterns. Recent decades have witnessed a surge in high-frequency trading (HFT), which has revolutionised market dynamics and drawn scholarly focus towards higher-frequency trading behaviours. This shift, marked by advancements in trading technology, has unveiled momentum effects within HFT and has prompted many scholars to study higher level frequency of trading behaviour in financial markets and find supportive evidence for the existence of momentum effects in high-frequency trading (e.g., Chaboud et al., 2014; Gao et al., 2018; Lou et al., 2019; Baltussen et al., 2021; Li et al., 2022). Among them, Gao et al. (2018) focus on intraday momentum in the US stock market and they show that market returns in the first half hour of the trading day can predict last half-hour returns before closing. This intraday momentum is quite strong and remains so even after accounting for reasonable transaction costs. Li et al. (2022) extend the

study of intraday momentum to international stock markets and find that this phenomenon is statistically and economically widespread in international markets. Their empirical results show that investment strategies based on intraday momentum can earn significant profitability across 12 developed markets; and that trading only in the last half-hour is more profitable with lower exposure risk than investing in the entire trading day.

Although Gao et al. (2018) and Li et al. (2022) provide evidence of intraday time-series momentum, they focus only on the momentum in a single market and ignore the linkages between markets that may lead to cross-market momentum. Extending the works of Gao et al. (2018) and Li et al. (2022), our study is the first to provide evidence on cross-market intraday momentum, making a unique contribution to the existing literature on intraday momentum. Our analysis reveals not only the presence of intraday momentum within individual stock market but also across markets. Specifically, we find that the US market's last half-hour return significantly predicts the next day's first half-hour return in international markets. This insight forms the basis of our cross-market intraday time-series momentum (CITSM) strategy, which leverages the US stock market's last half-hour return as the signal to trade at the next day's first half-hour in international markets, and the strategy is demonstrated to be profitable.

We explain the cross-market momentum effect through the lead-lag relationship among international stock markets. In an internationally integrated financial environment, certain indicators of the US stock market can be used as predictors of international stock market performance. Market anomalies or “shocks” in the US stock market can also be observed in other stock markets (Jegadeesh and Titman, 1993; Fama and French, 1996; Rouwenhorst, 1998; 1999; Grundy and Martin, 2001; Titman et al., 2004; Bali et al., 2011; Kelly and Jiang, 2014; Corradin and Maddaloni, 2020; Xu et al., 2023). Rapach et al. (2013) find that the US market has a leading role in the international stock market, and its stock market returns are a useful

predictor of other industrialised countries, and the performance of this predictor is even better than other markets' own economic indicators.

We investigate further the behavioural aspects underpinning cross-market momentum. Vayanos and Woolley (2013) posit that market shocks, often originating from sources extraneous to the stock market, compel cross-market investors to reconfigure their portfolios, thereby inducing momentum. This momentum may partly result from infrequent portfolio rebalancing by investors (Bogousslavsky, 2016). With the significant role of the US stock market, its influence on cross-market investors is substantial; many institutional investors place significant emphasis on the closing prices of the US stocks, particularly the last half-hour's trading, which are deemed crucial for trading decisions (Cushing and Madhavan, 2000; Foucault et al., 2005; Rapach et al., 2013; Baltussen et al., 2021). Consequently, cross-market investors often adjust their international stock portfolio allocations based on these US market closing insights, especially the last half-hour information, a likely outset of cross-market intraday momentum.

Previous research on the overnight effect in stock markets provides theoretical support for cross-market momentum. Despite higher price efficiency and more information disclosure during trading hours, overnight price changes, on average, contain more considerable information. Nonetheless, overnight trading is featured with poorer liquidity and higher costs, and the night market cannot fully digest the information; as a result, overnight information is usually assimilated and reflected in stock returns within the initial half-hour of market trading (Barclay & Hendershott, 2003; Gao et al., 2018; Lou et al., 2019). Due to the time zone differences and varied market opening times, the closing of US market typically precedes the opening of international markets. Given the significant role of the US stock market in the global financial system, its closing information becomes an important component of overnight

information for other international stock markets, thereby acting as a significant overnight influence on other markets upon their opening.

In this context, our empirical approach includes a series of tests to examine the cross-market intraday momentum. Specifically, our study is conducted in four steps. Initially, we employ the standard predictive regression methodology of Gao et al. (2018), conducting time-series regressions between the first half-hour returns of 15 developed markets and the preceding day's last half-hour returns of the US market. This analysis aims to provide a comprehensive measure of international market outcomes. The regression results indicate significant predictability, both in-sample and out-of-sample, with US market returns exhibiting substantial predictive power. We also find that the last half-hour returns of the US market have significant predictive power throughout the sample period, but they are stronger during recession periods, which is consistent with Gao et al. (2018).

Secondly, Li et al. (2022) assert that the statistically significant predictive power of intraday momentum can yield economic gains. Consequently, we develop a straightforward trading strategy, termed as the cross-market intraday time-series momentum (CITSM) strategy, which capitalises on this cross-market intraday momentum effect. The CITSM strategy utilises the last half-hour return of the US market as a signal to construct either a long or short portfolio in the first half-hour of own international market on the following day. This strategy yields significant positive returns in 12 markets and surpasses the passive buy-and-hold benchmark in terms of Sharpe ratios in 11 markets.

Thirdly, to delve deeper into the CITSM strategy's profitability, we conduct a battery of cross-sectional analyses on the strategy's profits in international markets. Our findings indicate that the profitability of the CITSM strategy correlates with some specific market characteristics. Notably, the CITSM strategy performs more robustly in international markets with lower

liquidity or higher information uncertainty. Furthermore, the strategy demonstrates strong profitability in international markets when the US market experiences high information uncertainty. Interestingly, it also exhibits strong profitability under conditions of extremely low or high liquidity in the US market; this may be related to the impact of information shock from the US market to international stock markets.

Finally, to confirm the robustness of the CITSM strategy's profitability, we extend the study of Gao et al. (2018) by employing exchange traded funds (ETFs) of tradable international stock market indices as proxies for market returns. We then re-construct the CITSM strategy, incorporating reasonable transaction costs in each trade to better simulate real investment scenarios. Our results demonstrate that the CITSM strategy continues to generate excess returns in over half of the markets even after accounting for such factors, thereby confirming its robust profitability in international stock markets.

Our paper further contributes to the research on cross-market momentum. Jostova et al. (2013) and Geczy and Samonov (2017) advocate that incorporating the influence of other markets, such as bonds, can enhance the performance of traditional single-market momentum strategies. They also provide supportive evidence in international contexts. Similarly, Chevallier and Ielpo (2014), along with Lim et al. (2018), demonstrate that strategies employing cross-asset dual momentum yield higher profitability compared to conventional single-market momentum approaches. Using a novel approach, Pitkäjärvi et al. (2020) integrate bond and stock markets in constructing their momentum portfolio, achieving a Sharpe ratio that surpasses traditional time-series momentum portfolios by 45%.

While these studies predominantly focus on the momentum interplay between financial assets like bonds and stock markets, and they typically concentrate on monthly or weekly frequency, our research diverges from them by examining higher-frequency data. We investigate the

intraday momentum relationship between different stock markets, drawing on the methodology of Rapach et al. (2013). Our findings reveal compelling evidence that the US market retains its leading position at an intraday level. Moreover, US market returns exhibit even greater predictive power than many individual market returns; intraday momentum strategies that utilise US returns as signals have also proven to be significantly profitable in international markets. This finding offers a substantial contribution to the existing literature on cross-market momentum, expanding our understanding of market dynamics at a higher frequency level.

The remainder of the paper is organised as follows. Section 2 describes the data used in the analysis. Section 3 specifies the models. Section 4 presents the results on cross-market intraday momentum in international stock markets. Section 5 discusses the economic significance of our findings. Section 6 discusses the impact of market characteristics on the CITSM. Section 7 presents the robustness checks. Section 8 concludes the paper.

2. Data selection and intraday return construction

2.1. Data selection

For our investigation into high-frequency stock market return predictability, we collect minute-level intraday data on international stock markets from the Refinitiv DataScope database, and then use it to construct half-hour stock returns for subsequent testing. Following the methodology of Li et al. (2022), we select 16 MSCI-classified developed market stock indices, including the US stock index, to examine the predictability. We posit that a comprehensive coverage of these developed markets can provide generalised evidence applicable to international markets. Additionally, for robustness checks, we also collect ETF data pertaining to these 16 markets from the same database. The sample date spans of these 16 MSCI developed markets vary; we use the longest available sample period for each market. Table 1

details the specific sample periods, including index names, trading hours, and other important information.

[Insert Table 1 about here]

Given our focus on the influence of US stocks on other markets, we use our stock market index data denominated in US dollar currency, rather than local currencies. This approach, which includes intraday trading time, volume, opening price, highest price, lowest price, and latest price per minute, mitigates potential issues related to exchange rate conversions and provides a unified standard for constructing intraday strategies in different markets. Consistent with Lawrenz and Zorn (2017) and Li et al. (2022), our regression results remain robust whether local or US dollar currency is used, indicating that exchange rate fluctuations have no significant impact on international stock market predictability.

Using high-frequency data often involves careful data cleaning, which is crucial for accurate stock estimation. We follow the high-frequency data cleaning procedures proposed by Barndorff-Nielsen et al. (2009) and Hollstein et al. (2020). We use only valid trading data with minute timestamps during intraday trading hours for each market. Due to market liquidity reasons or database recording mismatches, sometimes the stock market price remains unchanged a few minutes before the market closing, or there are still price changes after the market closing. To address these problems, we use the last actively changed price as the closing price. Additionally, given the vast volume of high-frequency trading data, we filter out extreme values by removing prices exceeding three standard deviations from the average volatility of the data over the entire sample period.

2.2 Intraday return construction

In high-frequency research, researchers usually divide each trading day into 30-minute non-overlapping intervals; the first half hour subsequent to the stock market's opening and the last half hour prior to the market's closing tend to have higher turnover, volume and volatility than other times throughout the trading day. These two periods are important for both high-frequency investors and financial researchers (Heston et al., 2010; Komarov, 2017; Gao et al. 2018; Li et al., 2022). We follow Li et al. (2022) to focus on first and last half-hour returns of 16 developed market indexes. The first and last half-hour returns are respectively defined as follows:

$$R_{F,t} = \frac{P_{F30,t}}{P_{Open,t}} - 1, \quad (1)$$

$$R_{L,t} = \frac{P_{Close,t}}{P_{L30,t}} - 1 \quad (2)$$

where $R_{F,t}$ denotes the first half-hour return on day t , $P_{Open,t}$ is the opening price on day t and $P_{F30,t}$ represents the last price in the first 30 minutes after the opening of day t . In the same vein, $R_{L,t}$ denotes the last half-hour return on day t , $P_{close,t}$ is the closing price and $P_{L30,t}$ represents the first price in the last 30 minutes before closing.

Our method diverges from previous research by omitting overnight information from the trading signal of the half-hour returns (e.g., Gao et al., 2018; Li et al., 2022). This is because we focus primarily on the impact of the last half-hour returns in the US market ($R_{L,t-1}^{US}$) on the first half-hour returns of the next day in other markets ($R_{F,t}^a$). While the time span of the overnight is long, we cannot obtain all overnight information before other markets open. Taking the US stock market as an example, the overnight time span between the closing of a trading day and the opening of the next day is 17.5 hours, and the intraday trading span is only 6.5 hours in comparison. The exclusion of overnight information is due to the complexities in defining the scope of such information across different international markets, especially

considering the time zone differences. Additionally, from a practical investment perspective, high-frequency traders typically avoid holding positions overnight, often closing or reducing their positions before the market closes to mitigate risk (Menkveld, 2016). Consequently, our analysis specifically targets intraday returns, offering a more focused insight into the day-to-day dynamics of global stock markets.

Figure 1 visualises and compares the trading hours measured as Coordinated Universal Time (UTC) for the US stock market and other 15 developed markets. The red columns show the trading hours of the US in the previous day, and with its last half hour column is particularly coloured dark red, reflecting the period during which the US market return will be used as a predictor of international stock markets in the study. Even after accounting for the effects of summer/winter time in each market, the previous day's trading hours for US stocks do not overlap with and lead the same day's trading hours for other markets. These non-overlapped trading hours provide opportunities for investors in these markets to earn extra gains by trading on the daily information of the US stock market.

[Insert Figure 1 about here]

Table 2 presents the descriptive statistics, including the number of days (No. Days), mean, standard deviation (SD), skewness and kurtosis for the first half-hour returns of 15 international stock market indices and the last half-hour return of US stock market index. It shows that most markets have positive mean returns with negative skewness, indicating that international stock markets generally have more days with positive return over the past few decades.

[Insert Table 2 about here]

3. Model specification

3.1 Time-series model

To examine the impact of the US market last half-hour return on the next day's first half-hour return of international stock markets, we conduct a series of regression tests. We first follow Gao et al. (2018) and run the following predictive regression:

$$R_{F,t}^i = \alpha + \beta_1 R_{L,t-1}^a + \beta_2 R_{L,t-1}^{US} + \varepsilon_t, \quad t = 1, 2, \dots, T \quad (3)$$

$$R_{F,t}^i = \alpha + \beta_1 R_{ON,t-1}^a + \beta_2 R_{L,t-1}^{US} + \varepsilon_t, \quad t = 1, 2, \dots, T \quad (4)$$

where $R_{F,t}^a$ and $R_{L,t-1}^a$ denote the first half-hour return and the last half-hour return for each non-US market in 15 international stock markets at day t and day $t-1$, respectively; $R_{ON,t-1}^a$ denotes the overnight return for each non-US market in 15 international stock markets from closing at day $t-1$ to opening at day t ; $R_{L,t-1}^{US}$ is the last half-hour return for US market in day $t-1$. T is the total number of trading days in the sample.

We are also interested in the predictive performance of the last-half hour return of the US market under different conditions, such as recession and non-recession periods. The regressions for these recession and non-recession periods are as follows:

$$R_{F,t}^a = \alpha + \delta_1 D_{t-1} \times R_{L,t-1}^{US} + \varepsilon_t \quad (5)$$

$$R_{F,t}^a = \alpha + \delta_1 ND_{t-1} \times R_{L,t-1}^{US} + \varepsilon_t \quad (6)$$

where D represents a dummy variable that indicates recession periods in international financial markets, and the dummy variable takes value “1” in the recession period and “0” otherwise; ND indicates non-recession periods, a dummy variable taking value “1” in the non-recession period and “0” otherwise. δ is the coefficient of the dummy variable. The recession and non-recession dummy variables are based on the Organisation for Economic Co-operation and Development (OECD) market recession indicator compiled by the National Bureau of Economic Research (NBER). The data are sourced from the Federal Reserve Economic Data (FRED) website.

Moreover, we follow Welch and Goyal (2008) and perform the out-of-sample test (OOS) to conduct the robustness check on the stock return predictability in international markets. We use the out-of-sample R -squared (R_{os}^2) statistic to measure the predictive performance.

$$R_{os}^2 = 1 - \frac{\sum_{t=n}^T (r_{t,L}^a - \hat{r}_{K,L}^a)^2}{\sum_{t=n}^T (r_{t,L}^a - \bar{r}_{K,L}^a)^2} \quad (7)$$

where R_{os}^2 is the out-of-sample R -squared; T is the total observation number and n is initial estimation period, counted in days; K is the out-of-sample observation number, counted by rolling window method or expanding window method; $r_{t,L}^a$ is the in-sample forecast of the first half-hour stock returns in day t ; $\hat{r}_{K,L}^a$ is the out-of-sample estimated value of stock returns from the previous K days up to day $t-1$, and $\bar{r}_{K,L}^a$ is the historical average of first half-hour stock returns. The model compares the mean squared prediction error (MSPE) with the historical value, with a positive R_{os}^2 implying a higher performance of indicator than the historical average level.

3.2 Cross-sectional model

To analyse the CITSM strategy in international stock markets, we follow Fama and MacBeth (1973) and test the cross-sectional relationship between the profitability of the CITSM strategy and 3 characteristic variables, including liquidity, information discreteness (ID), and volatility.

The corresponding cross-sectional regressions are as follows:

$$R_{S,t}^i = \alpha_t + \beta_t F_t^i + \epsilon_t^i \quad (8)$$

$$R_{S,t}^i = \alpha_t + \boldsymbol{\beta}_t' \mathbf{F}_t^i + \epsilon_t^i \quad (9)$$

where $R_{S,t}^i$ is the ITSM strategy return for market i at time t ; F_t^i is the characteristic variable for market i at time t . $\mathbf{F}_t^i$ is a 3-dimensional vector of all 3 characteristic variables for market i .

4. Cross-market intraday return predictability

4.1. Evidence of cross-market intraday momentum in international markets

We extend Gao et al. (2018) and Li et al. (2022) and investigate the relationship between the first and last half-hour returns of developed stock market indices. Differing from previous intraday studies, we focus on assessing the influence of the US market's last half-hour return on the first half-hour returns of international stock markets on the subsequent day. Choosing the first half-hour rather than other intraday trading periods for testing is because it is the first freely tradable period after overnight period, featuring higher liquidity and lower transaction costs, with a large concentration of institutional traders trading purely based on overnight information during this period. Conversely, trading at the close may reflect trades that are not purely information-based, and many of these trades are for rebalancing the non-optimal portfolios (Barclay & Hendershott, 2003; Gao et al., 2018; Lou et al., 2019). As previously mentioned, the last half-hour returns of the US stock market, as an important component of international stock market overnight information, are considered by institutional traders and reflected in the first half-hour of stock market trading, thereby theoretically having a predictable impact on stock market returns. We, therefore, construct corresponding regression tests to explore their connections. The respective 1-day lagged last half-hour returns and overnight returns of 15 international developed stock markets are respectively used as control variables in the regressions.

Table 3 presents the in-sample estimation results of our first predictive regressions. These results utilise both the US's last half-hour return and the respective last half-hour returns of 15 international developed stock markets as predictors for each market's first half-hour return on the following day. To address heteroscedasticity and autocorrelation issues, we apply the Newey-West (1987) correction method to adjust the standard errors of the regressions. The

figures in the second column of Table 3 reveal that only a small number of the 15 markets demonstrate in-sample predictability based on their own last half-hour returns. In contrast, the US market's last half-hour return, as shown in the third column, exhibits strong positive effects, with its predictive power evident in 11 markets and the coefficient estimates in 10 markets being statistically significant at the 1% level. We also consider the inter-market influences and estimate panel regression for the portfolio that including all the 15 markets. For the panel regression used in the tests, we use the random effects model with standard errors clustered by markets. As the results in the last row of Table 3, the US market's last half-hour return still shows significant coefficient estimates at the 1% level, consistent with the above separate market testing results. The market's own last half-hour return, although, performs poorly in separate market predictions, it demonstrates certain negative effects to the overall international markets. The results provide strong in-sample evidence that the last half-hour of the US market has a positive predictive effect on the international stock market returns in the first half-hour of the next day, and this predictive effect is much stronger than that of each market's own return at the last half-hour.

We now provide some reasons for the relatively weak predictive power of the last half-hour returns within their own markets. There is an overnight market that cannot be ignored (usually more than 14 hours) between the last half hour and the first half hour of the next day. The overnight period is relatively long, and investors are subjected to the uncertainty from economic or political events that may move the market; consequently, high-frequency investors are usually unwilling to hold positions overnight, but instead choose to close or reduce positions before the market closes to reduce overnight risks (Menkveld, 2016). As a result, the connection between the last and first half-hour returns is weakened, as they are separated by the overnight period. Night trading further complicates this dynamic, with information from the day market often being absorbed overnight rather than post-opening of the following day.

Literature suggests discernible patterns in stock market performance at night and differences in predictability between day and night trading sessions, with the US stock market displaying a positive premium overnight but close to zero or negative during the daytime (Miller, 1989; Cooper, Cliff, and Gulen, 2008; Kelly and Clark, 2011; Ye et al., 2023). Lou et al. (2019) split expected returns into overnight and intraday components, and find that contrary to the well-known aggregate effect, the high frequency trading has an economically and statistically significant overnight premium; almost all abnormal returns for momentum or reversal occur overnight, and this pattern is not driven by news.

Table 4 presents our second in-sample estimation results that utilise both the US's last half-hour return and the respective overnight returns of 15 international developed stock markets as predictors for each market's first half-hour return on the following day. It can be seen that both the US's last half-hour return and own overnight return show strong positive effects in most markets, suggesting that they are both good predictors of the international market's first half-hour returns. The significant coefficient of overnight return, however, disappears in the panel regression, indicating that its predictive power is not robust after considering the connection between the international stock markets. While the US's last half-hour return exhibits significant coefficient either in separate markets or in the entire international market, demonstrating robust predictive power that is better than overnight return.

[Insert Table 3 and Table 4 about here]

4.2. Evidence of cross-market intraday momentum under specific conditions

We now examine the relationship between the US market's first half-hour return and an international stock market's first half-hour return under specific market conditions, including recession and non-recession periods in economic cycles. Table 5 (both Panels A and B) shows that the last half-hour of US market demonstrates significant results in most international

markets, which indicates a strong predictive ability during both recession and non-recession periods.

Our analysis reveals a more powerful predictive power during recessionary periods, with 10 out of 15 markets showing significant positive slope coefficients during recessions (9 significant coefficient estimates in non-recession periods) and the adjusted R^2 is notably higher in recessionary periods. This finding aligns with previous research, indicating that the relationship between the first and last half-hour returns of the US market is more pronounced during economic recession periods (Gao et al., 2018). Many empirical studies suggest that not only high-frequency stock indicators but also a variety of daily and monthly macroeconomic indicators generally exhibit more predictive power during recessionary periods, especially during financial crises (e.g., Rapach et al., 2013; Bena et al., 2022; Borup & Schütte, 2022).

[Insert Table 5 about here]

4.3. Out-of-sample evidence of cross-market intraday momentum

Our previous regression analyses are based on in-sample estimates over the whole sample period, yielding the strong predictive power of the US market's last half-hour returns for the first half-hour returns of the next day in international markets. It is essential to determine whether the parameters of our regression model remain robust over time, especially considering the often-noted instability of macroeconomic predictive variables despite significant in-sample results (Ashley et al., 1980; Welch and Goyal, 2008). We thus investigate the out-of-sample (OOS) predictive power of the US market's last half-hour returns to assess whether the parameters of the regression are robust over time. We employ two methods, rolling window and expanding window, and use the first two years sample of each market as the initial sample estimation. The out-of-sample predictive power of predictor is then tested in two ways by

calculating the R_{OS}^2 and the MSPE-adjusted R_{OS}^2 (Clark and West, 2007; Campbell and Thompson, 2008).

The results of our out-of-sample tests, presented in Table 6, include R_{OS}^2 and corresponding p -values for each international market. Our findings indicate that 9 markets exhibit significant positive OOS predictability using the rolling window estimation method, and this number increases to 10 markets when considering the MSPE-adjusted OOS. These results underscore the robust predictive power of the US market's last half-hour returns across international markets. Though the expanding window estimates yield comparatively weaker results, they still demonstrate significant positive OOS predictability in 7 markets and MSPE-adjusted positive predictability in 10 markets. This consistency across different testing methods indicates that the US market's last half-hour returns provide additional predictive information relative to a simple historical average of the first half-hour returns in international markets. Overall, our OOS analysis confirms the predictive power of the US market's last half-hour return.

[Insert Table 6 about here]

5. Economic significance of cross-market intraday momentum

Economic benefits and statistical significance are often closely related. There is evidence that some indicators can produce profits in a portfolio context, even if they show relatively weak statistical predictive power (e.g., Cenesizoglu and Timmermann, 2012; Li et al., 2022). One way for assessing the performance of the indicator is to examine its performance in terms of market timing selection (Kandel and Stambaugh, 1996; Cenesizoglu and Timmermann, 2012; Gao et al., 2018). The economic significance of the cross-market intraday momentum, as highlighted by our research, extends beyond mere statistical relevance. The predictive power of the US market's last half-hour returns, as evidenced in our empirical results, translates into

tangible financial gains in international markets. To quantify this economic impact, we thus construct a CITSM strategy to conduct back-testing in international stock. The CITSM strategy uses the last half-hour return of the US stock market (R_L^{US}) as a timing signal for trading in international markets on the first half-hour in the next day. Specifically, if the timing signal from the US last half-hour is positive, we create a long position at the opening of the international stock market; alternatively, we create a short position at the market open. We then close the position at half an hour after the opening. The expression for the CITSM strategy is as follows:

$$CITSM(R_{F,t}^a) = \begin{cases} R_{F,t}^i, & \text{if } \text{sign}(R_{L,t-1}^{US}) > 0; \\ -R_{F,t}^i, & \text{if } \text{sign}(R_{L,t-1}^{US}) \leq 0. \end{cases} \quad (10)$$

where $CITSM(R_{F,t}^i)$ is the first half-hour investing profit ($\pm R_{F,t}^i$) of international stock markets on day t based on the US market's last half-hour return signal $\text{sign}(R_{L,t-1}^{US})$ on day $t-1$. Moreover, we also construct the naïve buy-and-hold strategy (BH) and always long strategy (AL) as the benchmarks to compare the profitability of the strategies. The BH strategy longs the stock index in each market at the beginning of the sample period and then holds the index until the period ends. The AL strategy always create a long position at the opening of the international stock market in each trading day and then close the position at half an hour after the opening.

Table 7 provides the annualised average returns, skewness, kurtosis, and Sharpe ratios of the CITSM strategy; the buy-and-hold strategy (BH) and the always long strategy (AL) in 15 stock markets, as well as the information ratios (IR) by using the BH and AL strategy respectively as the benchmark in each market. The results demonstrate that the CITSM strategy yields positive average annualised returns in 13 of the 15 markets throughout the sample period, outperforming the BH strategy in 11 of these markets and the AL strategy in 12 markets. Notably, the CITSM strategy has impressive profitability surpassing 20% in the Australian

stock market and Austrian stock market. This high profitability aligns closely with those markets that exhibit significant correlations with US returns in our regression analyses, affirming the notion that statistical significance in stock markets can indeed translate into tangible economic benefits. We also observe positive Sharpe ratios for the CITSM strategy in 12 markets, and that the IRs benchmarked to the BH strategy are positive in 11 of them (Meanwhile the IRs benchmarked to the AL strategy are positive in all 12 markets), indicating that the earning performance of the BH strategy and AL strategy are insignificant compared to the CITSM strategy in most international markets. Moreover, the skewness of the CITSM strategy return is positive in all 15 markets, revealing a low crash risk using the strategy in investment, which is favourable to investors. In contrast, the BH strategy and AL strategy deliver returns with negative skewness, showing higher risk in almost all markets investments. To display the profit performance of the strategies more intuitively, we also conduct alpha tests. Specifically, we use the daily profits of the BH strategy and the AL strategy as benchmarks respectively to examine the excess return (alpha) of the CITSM strategy relative to the benchmark strategy. The alpha test results are shown respectively in Panels A and B in Table 8. The table shows that the CITSM strategy is statistically significant in 12 out of 15 stock markets, and its alpha exceeds 10% in these markets. Among them, the alpha of the CITSM strategy in the Australian stock market even surpasses 88%, which means that the CITSM strategy can earn more than 88% excess return on average in each trading day compared to the BH and AL strategies, demonstrating significant profit performance. Overall, the results of the alpha test validate the significant outcomes presented in Table 7, confirming that the CITSM strategy possesses robust profitability and outperforms the BH strategy and AL strategy in the international stock markets.

[Insert Table 7 and Table 8 about here]

6. Cross-market intraday momentum – Cross-sectional analysis

In the preceding section, we have demonstrated the significant profitability of the CITSM strategy in international stock markets. To delve deeper into the determinants of CITSM's excess returns across these markets, we link the CITSM's performance to market microstructure, especially some specific market characteristics and conduct cross-sectional analyses. Following Li et al. (2022), we construct three market characteristic variables for each market, encompassing intraday liquidity, information discreteness (ID), and information uncertainty. We use volatility and spread as proxies for intraday liquidity and information uncertainty, respectively. The detailed construction process for all three characteristic variables can be found in the Appendix. The characteristic variables are plugged and used for Fama-MacBeth regression testing, and we also use two cross-sectional analyses based on characteristic variables, namely cross-market sorting and time-series sorting, to explain strategy returns from different perspectives.

6.1 Fama-MacBeth analysis

To study the cross-sectional relationship between CITSM profitability and characteristic variables, we first use Fama and MacBeth (1973) regression analysis on the entire international stock market. Table 9 reports the average annualised coefficient estimates from the Fama-MacBeth regression, the t -statistic adjusted by Newey and West (1987) standard errors, and the adjusted R^2 . According to the results, each characteristic variable exhibits significant cross-sectional relationship with the CITSM return. Volatility and spread, in particular, show significant results at the 1% level with large coefficients (4.73 and -3.95, respectively) and high R^2 values (15.81% and 17.68%). ID, while significant at the 10% level, appears to have a relatively minor impact on the strategy. When all three variables are included in the regression, their combined significance remains, and the R^2 reaches 36.45%, indicating that these variables

can explain more than 30% of the CITSM strategy's excess returns. Our preliminary evidence indicates that when the international stock market liquidity is lower, information dispersion is smaller; or when information uncertainty is higher, the CITSM strategy can achieve higher returns.

[Insert Table 9 about here]

6.2 Cross-market sorting analysis

To further understand the dynamics driving the CITSM strategy's profitability, we employ a cross-market sorting analysis. Specifically, we sort all the 15 markets according to the characteristic variables in each trading day, and daily split all markets into three groups, including the small group (bottom 30%), medium group (middle 40%) and large group (top 30%). Table 10 shows the annualised average return (AVE) and various statistical indicators, including standard deviation (SD), Sharpe ratio, Skewness and Kurtosis, for investment portfolios of 3 characteristic variables. We also report AVE and statistical indicators for each long-short portfolio (L-S) in the table.

It can be seen that the CITSM presents some obvious patterns in international stock markets investment. With the volatility of the market increases, the AVE increases from 0.58% to 12.92%; the L-S portfolio of volatility group also can earn 12.33%, demonstrating significant profitability. The impact of changes in spread has the opposite trend. The strategy can obtain a return of 11.14% when the spread is low, and earn 5.48% when the spread is high; the short-long portfolio (S-L) can obtain AVE of 5.66%. Although ID also shows a negative trend as the spread, the gap of AVE between the small, medium and large groups of ID is small, and its L-S portfolio return is insignificant as well, so the impact of ID on CITSM is relatively weaker compared to the other two characteristic variables. Overall, the results from the cross-market sorting regressions largely confirm our Fama-MacBeth regression results.

[Insert Table 10 about here]

6.3 Time-series sorting analysis

After conducting cross-market sorting analysis and discovering the obvious patterns between profitability of the CITSM strategy and market characteristics in international stock markets, we next further explore the unique impact of the characteristic variables in each market. We, therefore, construct a time-series sorting to sort trading days by the characteristics in each market and analyse the time-varying impact of the characteristic variables on the strategy returns.

Since the CITSM strategy uses the US market's last half-hour return to predict the first half-hour return of the international stock market at the next day, when doing the time-series sorting, we not only consider the own characteristic variables of each market, but also the impact of the US market's characteristic variables on strategy returns. Specifically, we first use the 3 characteristic variables' information of the US market in the last day to sort all trading days of each of the 15 markets and divide them into three groups (small, medium and large), and construct an equally-weighted CITSM investment portfolio in each group. We also use each market's own characteristic variable information to sort and group again, and also construct equally-weighted portfolios.

Table 11 reports the test results sorted by US market characteristics (Panel A) and own market characteristics (Panel B) respectively. It shows the annualised average returns (AVE) of each investment portfolio and each L-S portfolio. We compare the results of Panel A and Panel B. The results in the first column show that whether grouping is based on ID of the US market or its own market, there is no obvious pattern. The AVE of the small, medium and large groups in the 15 markets are not much different, and the returns of L-S portfolio of 15 markets are lower and are not consistent with the mixture of positive and negative results, indicating that

ID has a small impact on the CITSM.

Volatility shows more significant results in the second column. In the large Volatility group of the US market in Panel A, the CITSM has significant high AVE in 13 of the 15 markets and shows a significant return gap with the medium and small volatility group; its L-S portfolio also has positive AVE in all markets, indicating an obvious profit pattern. This pattern is particularly impressive in the Australian stock market, with 43% AVE in the high volatility group, and its L-S portfolio also earning 35%. One possible reason for this situation is that the Australian stock market has many companies with US capital holdings or financing, especially in the technology field. These companies dominate the market to a certain extent and are very sensitive to the impact from the US market, resulting in an obvious similar trend of the Australian market to the US stock market.

The volatility of own market also shows the same pattern in Panel B, and 12 markets still have significantly high AVE in the high volatility groups. The Australian stock market even reaches an astonishing 60% AVE, and the L-S investment portfolio reaches 57%, demonstrating high performance. Our results show that the CITSM strategy has stronger profitability when the market uncertainty is high in both the US market and international markets. This evidence is consistent with the view of Gao et al. (2018) and Li et al. (2022) that intraday momentum is highly correlated with market volatility, and momentum strategy returns are higher under severe market conditions.

It is worth mentioning that, unlike other characteristic variables, there is a different pattern between the US market spread and own market spread groups. We first look at Panel B; more than half markets shows higher return in the small group with the significant negative L-S returns, which illustrates a strong negative pattern. This result is consistent with the above cross-market sorting results. However, in the US market spread characteristic grouping of Panel A, the strategy exhibits high returns in both the low spread and high spread groups of

most markets, indicating that the CITSM strategy is profitable in international stock markets under both high and low intraday liquidity conditions in the US market. A possible explanation for this situation is that, as the US market is the largest and the most important trading market in the world, the extreme US high liquidity and low liquidity are both important information shocks to the international markets. These US shocks spread around the world and affect the international markets, making international stock market returns predictable; and the CITSM strategy thus gets significant profitability in the international stock markets investment.

[Insert Table 11 about here]

In addition, we also consider the joint impact of both the US market characteristics and own market characteristics on the CITSM strategy's profitability and therefore conduct double sorting analysis (the double-sorting test results are not reported for brevity but are available upon request). The results are similar to the above single sorting tests, the strategy exhibit higher return when the volatility of both US and own markets are high; when US spread is low and own market spread is high, the strategy can earn more profits.

7. Robustness Check

To assess the effectiveness of the CITSM strategy more accurately in a real-world context, we utilise ETFs associated with various market indices to gauge market returns. ETFs offer a preferable alternative to direct stock index investments due to their higher liquidity, reduced capital requirements, and near-equivalent representation of stock index movements. For instance, the SPY ETF, which mirrors the S&P 500 Index, demonstrates a minimal tracking error of -0.04% since its inception and boasts a low commission rate of 0.095%. This makes it an attractive, cost-efficient option for investors seeking exposure to the broad stock markets.

Our methodology involves substituting the active ETFs of each market index for the indices themselves within our investment framework. We gather the bid and ask prices at the opening

and 30 minutes of each trading day, and then use the difference between the opening ask (bid) price and the first half hour bid (ask) price to calculate the long (short) position profit on that day. The ETF-CITSM strategy is formulated around this concept, which uses the US stock index ETF's (SPY) last half-hour returns before closing as the timing signal to invest in ETFs of international stock markets.

Perold (1988) highlights a common disparity between theoretical portfolio returns and the actual returns realised through live trading. To better align with real-world trading conditions, our study incorporates commission and fee costs into each trading transaction. Given the nature of high-frequency trading, which entails frequent positional adjustments, transaction costs can markedly influence the return performance of intraday strategies. Thus, a careful consideration of these costs is essential. Following the approach by Gao et al. (2018), we adopt a varied fee structure: a higher fee of \$10 per trade for casual traders and a lower fee of \$2 per trade for active traders when transaction sizes exceed \$10,000, in addition to a moderate transaction fee of \$6 per trade.

In our examination of the ETF-CITSM strategy's viability across different phases of the economic cycle, we aim to determine its consistent profitability rather than its performance during specific economic periods. Consequently, we omit two stock index ETFs— Ireland (EIRL) and Portugal (PGAL)—from our sample due to their lack of data spanning of at least ten years. The findings pertaining to the remaining 13 ETF indices are presented in Table 12.

[Insert Table 12 about here]

Table 12 reports the average annual returns of ETF-CITSM in the international stock markets and incorporate 4 kinds of transaction costs scenarios including no cost (NC), low cost of 0.02% per trade (LC), medium cost of 0.06% per trade (MC), and high cost of 0.10% per trade (HC). It shows that the ETF-CITSM strategy delivers positive returns and exceeds the BH one in all 13 markets during the sample period. This performance surpasses even that of direct

investments in stock indices. In addition, for the markets with higher returns shown in Table 7 (such as Australia, Austria, and Japan), their ETFs can still achieve higher returns than other markets, thereby underscoring the robust profitability of the CITSM strategy.

The impact of different transaction costs on the strategy's returns is also detailed in Table 12. With transaction costs of 0.02% for active traders, the returns of ETF-CITSM remain significant across all markets. When the cost increases to a moderate 0.06%, the strategy's return diminishes but remains positive in 8 markets, with 5 outperforming the BH strategy. However, for casual traders facing transaction costs of 0.1%, the profitability of the ETF-CITSM strategy is considerably affected; only in 3 markets (Australia, Austria, Japan) the strategy delivers positive returns and the performance exceeds that of the BH strategy. These 3 markets consistently exhibit high returns, regardless of whether one invests directly in stock indices or their corresponding ETFs, highlighting the exceptional profitability of the CITSM in these markets.

It is worthy to note that while transaction costs have a noticeable impact on returns, our strategy involves daily trading, which should realistically incur lower costs than those estimated for casual traders. Our analysis reveals that intraday momentum profits in cross markets persist after accounting for appropriate transaction costs.

8. Conclusion

Gao et al. (2018) and Li et al. (2022) introduce intraday time-series momentum (ITSM), and they demonstrate that the first half-hour return significantly predicts the last half-hour return in international stock markets. Our study takes into consideration the well-documented linkages between stock markets and introduces new cross-market intraday time-series momentum (CITSM) strategy.

Our study first confirms the leading role of the US market in influencing the international

markets at intraday level, finding that the US market's last half hour returns can predict the international markets' first half-hour returns at the next day. This phenomenon is both statistically and economically significant around the world; the intraday trading based on this cross-market momentum effect generates considerable economic benefits in terms of investment timing and asset allocation in international stock markets. Specifically, the returns of the majority of the developed markets in our sample show clear evidence of intraday predictability both in-sample and out-of-sample. The CITSM strategy, which utilises the last half-hour returns of the US stock market as a signal for constructing portfolios in international markets, outperforms the passive buy-and-hold strategy and always long strategy in terms of Sharpe ratios in 11 out of 15 markets, indicating its significant profitability. This profitability still remains substantial even after accounting for appropriate transaction costs.

Our further exploration into the profitability of the CITSM strategy in international markets reveals that its profitability is tied to some specific market characteristics. Our research indicates that CITSM's profitability is enhanced in scenarios where international market liquidity is low or information uncertainty is high. Additionally, the CITSM strategy also demonstrates strong profitability in conditions where the US market experiences high information uncertainty, or when the US market's liquidity is at extreme levels, either low or high.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 4 only for the improvement of the language and readability of the manuscript. After using this tool/service, the authors have carefully reviewed and edited the content as needed and assume full responsibility for the content of the publication.

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Appendix

Following Li et al. (2022), we construct 3 market characteristic variables, including intraday liquidity, information discreteness (ID), and information uncertainty. We use volatility and spread as proxies for intraday liquidity and information uncertainty, respectively.

We define intraday volatility as follows:

$$\sigma_t^2 = \frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2,$$

$$\bar{r}_t = \frac{1}{n} \sum_{i=1}^n r_i$$

where σ_t^2 denotes the last half-hour intraday volatility measure on day t . r_i presents the minutely return in minute i , $i = 1, 2, \dots, n$, and n is the number of minutes in the last half-hour for which have valid data. $\bar{r}_t$ denotes the average return of the last half-hour on trading day t .

The information discreteness (ID) is defined following Lim et al. (2018), shown as follows:

$$ID_t = \text{sign}(r_t^F) \times (\%neg_t - \%pos_t)$$

where r_t^F is the last half-hour return on day t ; $\%neg_t$ and $\%pos_t$ are the percentage of valid minute data associated with a negative and positive return within the last half-hour, respectively, on day t . A smaller ID means that information is absorbed relatively smoothly, while a larger ID means that the information is more discrete in the market.

To evaluate information uncertainty, we employ the percent-cost high-low liquidity measure as outlined by Corwin and Schultz (2012) and further refined by Li et al. (2022). This approach utilises the high and low prices from two adjacent time periods to estimate the percentage spread. The high-low liquidity is computed as follows:

$$S = \frac{2(e^\alpha - 1)}{1 + e^\alpha},$$

$$\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}},$$

$$\beta = \sum_{j=0}^1 \left[\ln \left(\frac{H_{t+j}}{L_{t+j}} \right) \right]^2,$$

$$\gamma = \left[\ln \left(\frac{H_{t+j}}{L_{t+j}} \right) \right]^2$$

where S denotes the intraday high-low liquidity measure, and H_t and L_t represent the high and low prices at time t respectively; $H_{t,t+1}$ and $L_{t,t+1}$ are the high price and the low price over two consecutive time periods t and $t + 1$. For each market, we estimate the spread in the last half-hour by averaging estimates across overlapping 5-minute intervals. Specifically, the high-low liquidity measure is calculated for every two consecutive 5-minute intervals, and the average is taken across these overlapping intervals within the last half-hour.